

Air Quality Prediction

Problem Statement

Air pollution is one of the major environmental challenges affecting public health and climate. The objective of this project is to develop a Machine Learning model that predicts the Air Quality Index (AQI) based on environmental factors such as PM2.5, PM10, temperature, location, and time-based features. Accurate AQI prediction enables early warnings, better environmental planning, and data-driven decisions.

Client Benefits

- Early pollution alerts and preventive action
- Improved public health protection
- Better urban and environmental planning
- Data-driven decision making
- Efficient allocation of pollution control resources
- Understanding seasonal and weekly pollution trends
- Smart city sustainability integration
- Long-term environmental forecasting

Dataset Description

The dataset contains environmental data used to predict AQI. Key features include location, PM2.5, PM10, temperature, and date. The target variable is predicted_aqi, which is continuous, making this a regression problem.

Final dataset size: 1460 rows and 7 columns.

Data Cleaning and Preprocessing

- Converted date column into datetime format.
- Checked and handled missing values.
- Removed unnecessary columns.
- Verified data types and duplicates.

Feature Engineering

Extracted month and day_of_week from the date column to capture seasonal and weekly pollution patterns. The original date column was dropped to avoid redundancy.

Exploratory Data Analysis (EDA)

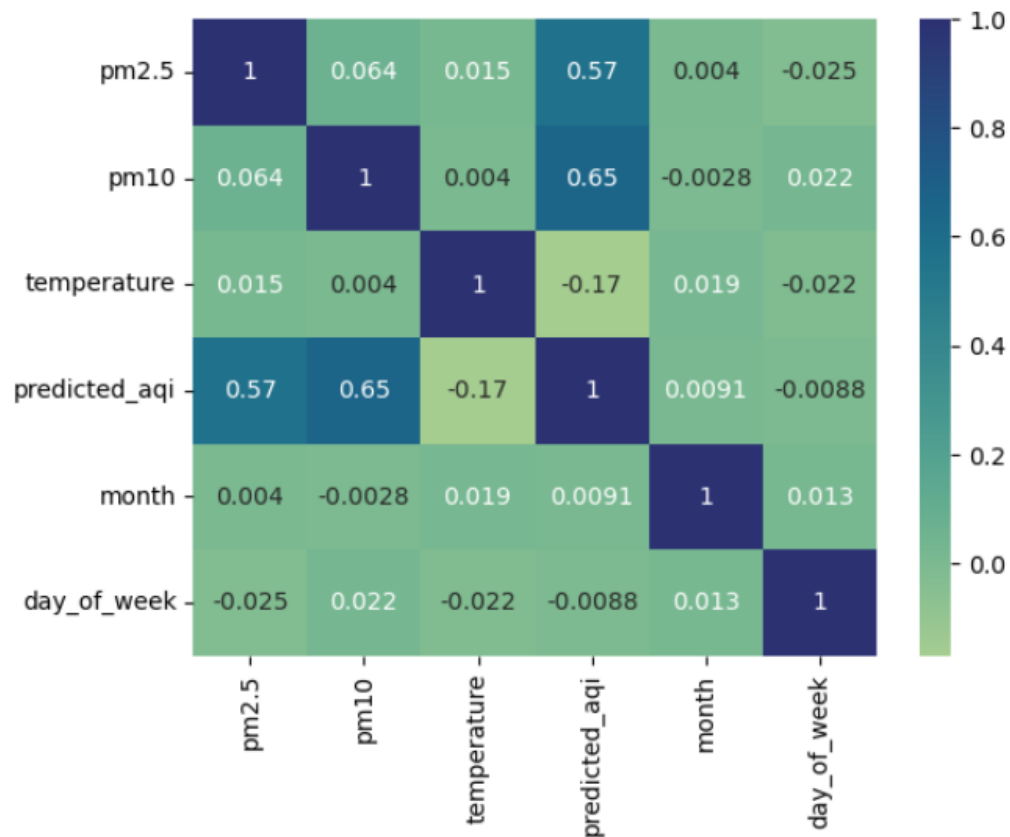
Performed correlation analysis and distribution checks to understand relationships between pollutants and AQI. Boxplots were used to detect outliers.

Graph Placeholders

Figure 1: Correlation Heatmap between features and AQI

```
sns.heatmap(aqp[['pm2.5', 'pm10', 'temperature', 'predicted_aqi', 'month',  
                'day_of_week']].corr(),annot=True,cmap="crest")
```

<Axes: >

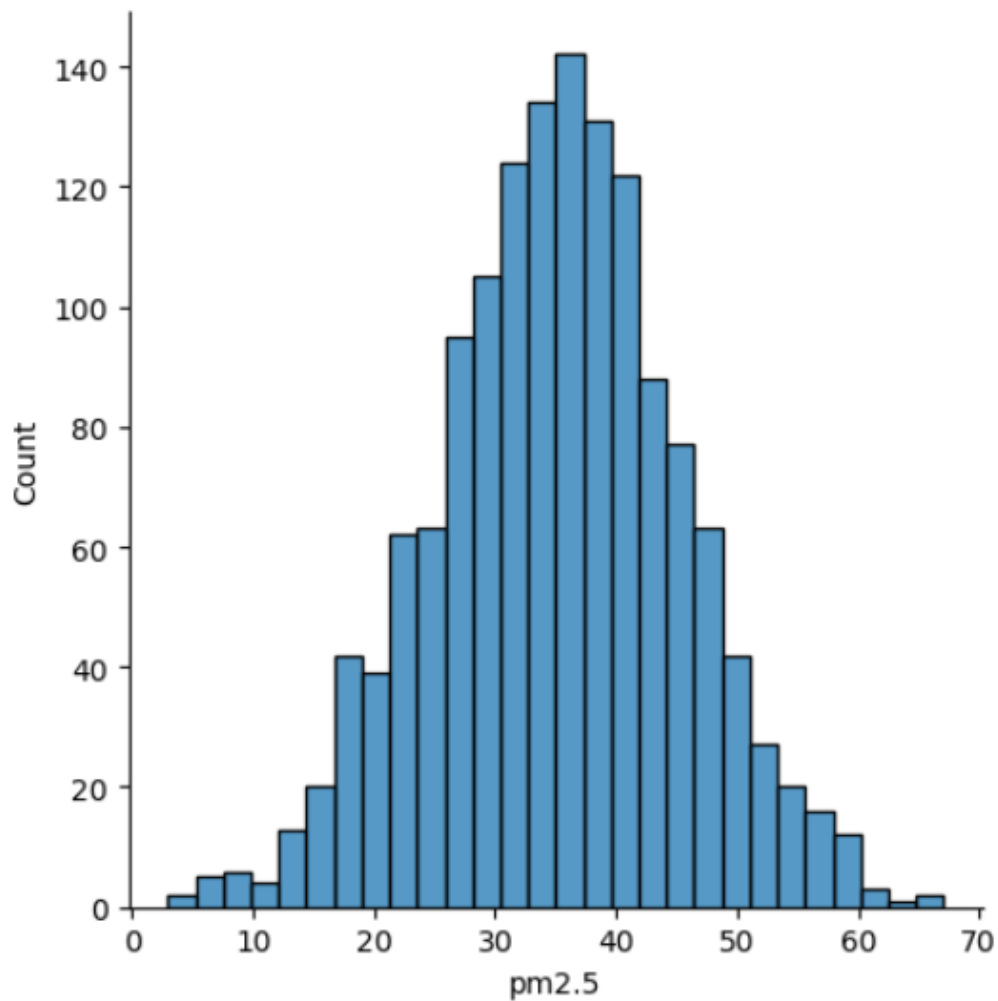


The heatmap shows that PM10 (0.65) and PM2.5 (0.57) have a strong positive correlation with AQI, indicating they are key predictors. Temperature has a weak negative correlation, while month and day_of_week have minimal impact.

Figure 2: Distribution plot of PM2.5

```
sns.displot(aqp, x="pm2.5")
```

```
<seaborn.axisgrid.FacetGrid at 0x2924a937a10>
```

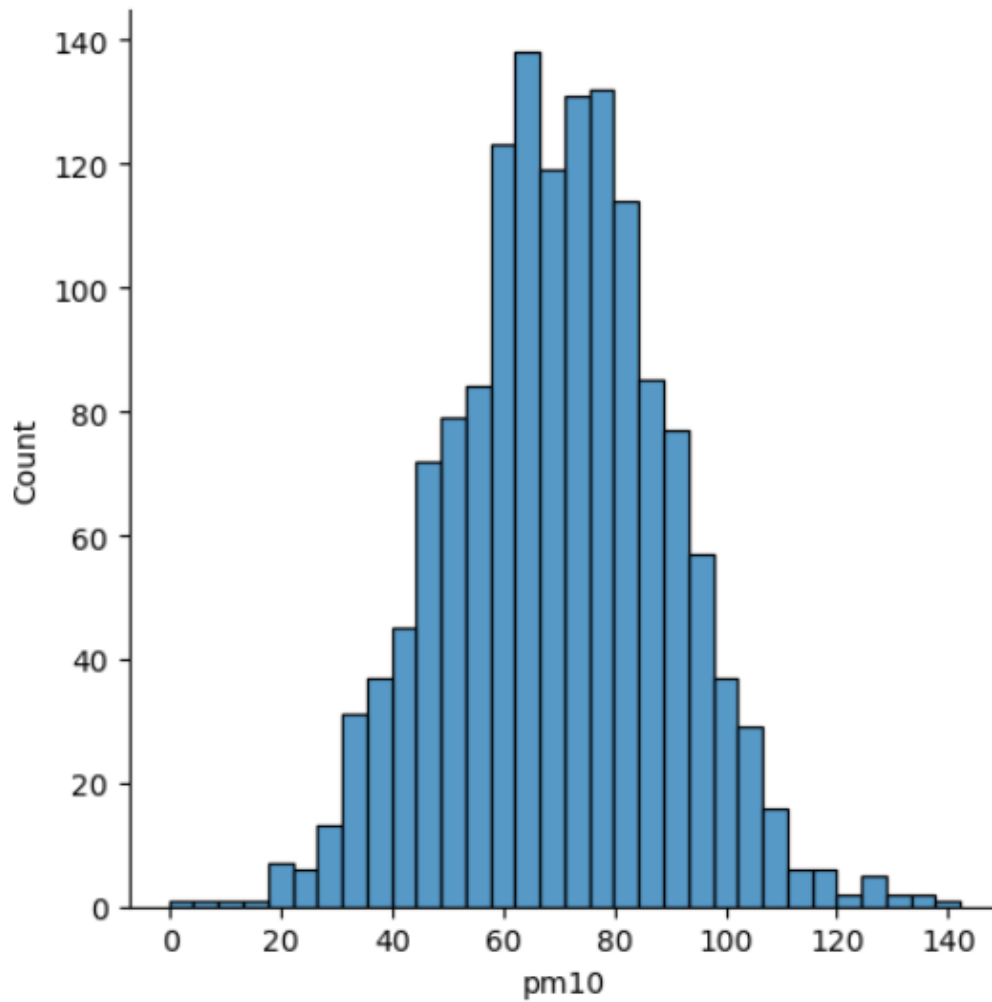


The PM2.5 values follow an approximately normal distribution with most values concentrated between 25 and 45. The peak occurs around 35–40, and no significant outliers are observed.

Figure 3: Distribution plot of PM10

```
sns.displot(aqp, x="pm10")
```

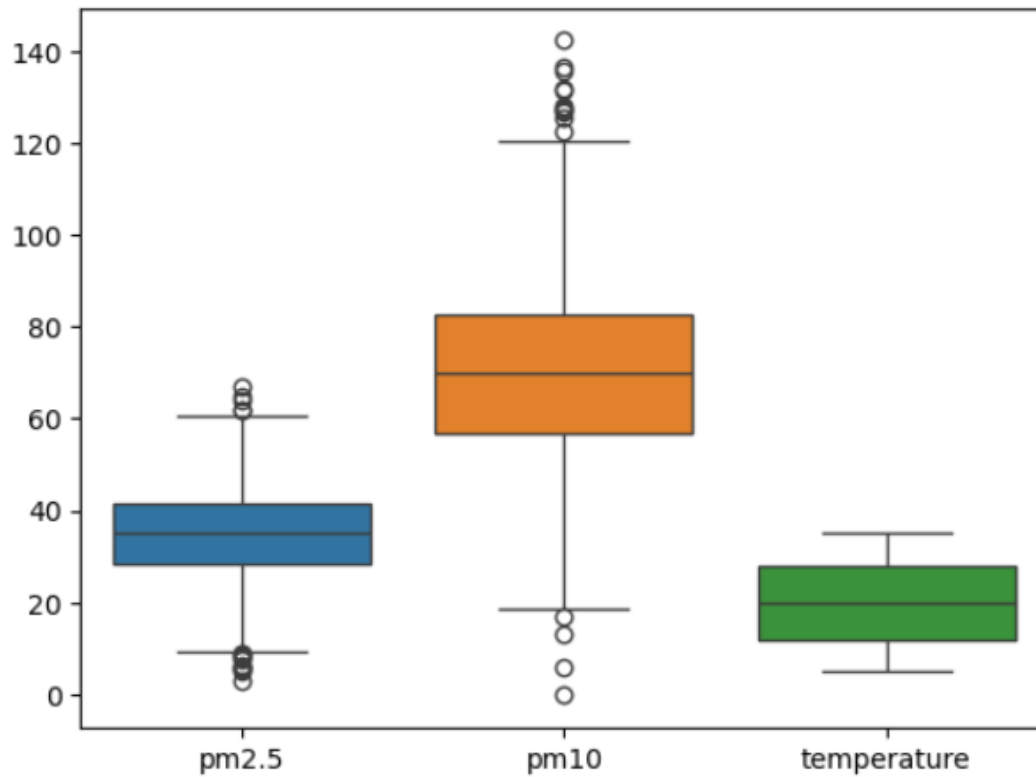
```
<seaborn.axisgrid.FacetGrid at 0x2924ab2f9d0>
```



The PM10 values show an approximately normal distribution, with most values between 50 and 90 and a peak around 65–75. The data does not show significant skewness or extreme outliers.

Figure 4: Boxplot of AQI

```
for i in num_cols:  
    sns.boxplot(aqp[i])  
    plt.show()
```

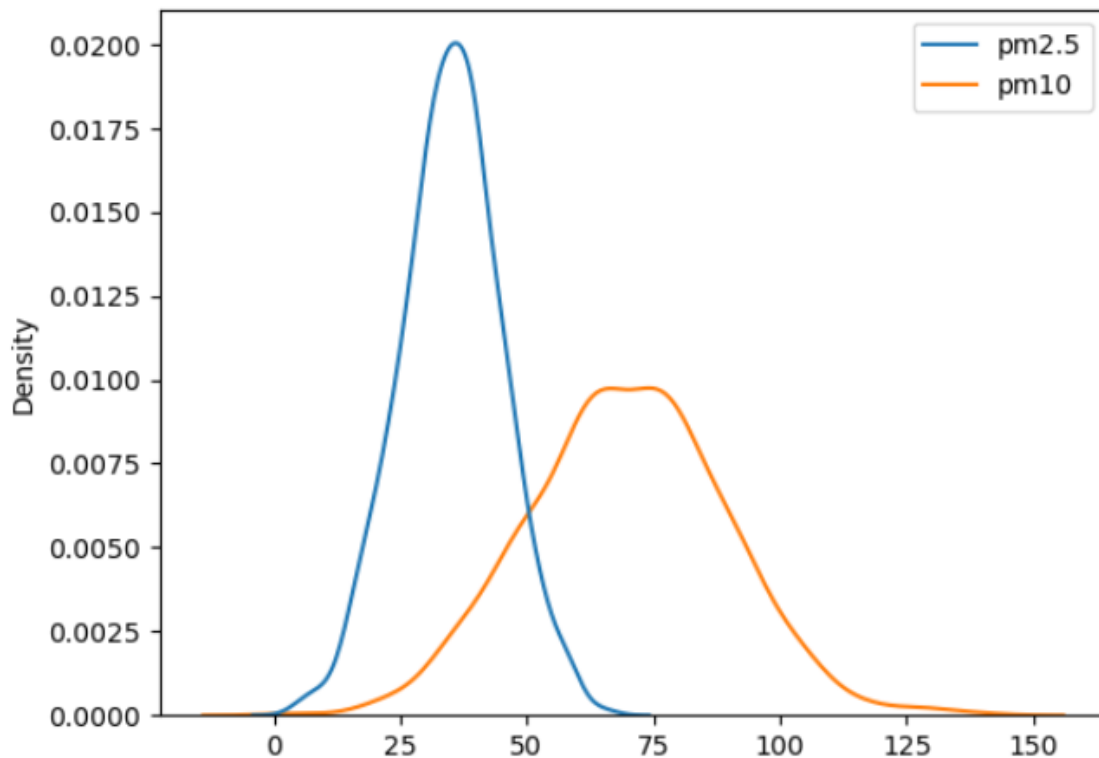


Boxplots revealed the presence of extreme values in pollutant variables such as PM2.5 and PM10. These outliers correspond to real pollution events and were therefore retained to preserve real-world data characteristics.

Figure 5: KDE Plot of PM2.5 and PM10

```
sns.kdeplot(aqp[["pm2.5","pm10"]])
```

<Axes: ylabel='Density'>



The KDE plot shows that PM2.5 and PM10 follow smooth, unimodal distributions. While both variables approximately resemble a normal distribution, they exhibit positive skewness, with PM10 showing greater variability and deviation from normality than PM2.5.

Feature Scaling

Applied StandardScaler to normalize feature values for models sensitive to feature magnitude. Tree-based models were trained without scaling.

Regression Models Applied

- Linear Regression
- Polynomial Regression (Degree = 2)
- Decision Tree Regressor
- Random Forest Regressor

Linear Regression Results

- The **R² score of 0.7747** indicates that the model explains approximately **77.47% of the variance** in the AQI.
- The **MSE value of 23.81** represents the average squared difference between actual and predicted AQI values.

- Cross-validation with 5 folds produced scores ranging between 0.703 and 0.774, with an average R^2 of 0.731.
- This shows that while the single train-test split gave a slightly higher testing score (0.775), cross-validation provides a more reliable estimate of the model's generalization ability, averaging around 0.731.
- Overall, the model is consistent, but cross-validation highlights that its true performance is slightly lower than the single test split suggests.

Polynomial Regression Results

After applying Polynomial Features (degree=2), the feature size increased to 28 features. Model Performance:

- Mean Squared Error (MSE): 24.50
- R^2 Score: 0.768

Cross-validation confirms that degree 2 polynomial regression generalizes best, with an average R^2 of ~ 0.73 , closely matching the single split score. Higher degrees lead to severe overfitting and poor generalization

The model explains approximately 76.8% variance in AQI.

Decision Tree Regressor Results

The large gap between the training score (1.0) and testing score (0.49) indicates that the Decision Tree Regressor is overfitting the training data. The model captures noise and extreme patterns present in the training set, which reduces its ability to generalize new data. GridSearchCV was applied to optimize the Decision Tree Regressor by tuning hyperparameters such as max_depth, min_samples_split, min_samples_leaf, and criterion. The best estimator obtained from GridSearchCV was then trained and evaluated on the test dataset. After hyperparameter tuning, the training score reduced to approximately 0.72 and the testing score improved to approximately 0.69. This reduction in training performance along with improvement in testing performance indicates a significant reduction in overfitting.

Random Forest Regressor Results

The Random Forest Regressor achieved a training score of approximately 0.95 and a testing score of approximately 0.71. The close alignment between training and testing scores indicates good generalization performance with minimal overfitting. The Mean Squared Error (MSE) value of approximately 30.24 shows reduced prediction error compared to single Decision Tree models. The R^2 score of 0.71 indicates that the model explains about 71% of the variance in the target variable.

Model Evaluation

Regression models were evaluated using:

- Mean Squared Error (MSE) – Measures average squared prediction error.
- R^2 Score – Measures proportion of variance explained by the model.

Model Comparison Summary

- **Linear Regression:** Best balance of accuracy and simplicity. It achieves the highest test R^2 (0.7747) and lowest MSE (23.81), making it the most reliable for AQI prediction in this dataset.
- **Polynomial Regression:** Adds complexity without improving performance. Slightly lower R^2 and higher MSE than linear regression.
- **Decision Tree (tuned):** Improved generalization after hyperparameter tuning, but still weaker than linear regression in terms of test R^2 .
- **Random Forest:** More robust than a single tree, but its test R^2 (0.72) and MSE (30.24) are worse than linear regression. It avoids overfitting but sacrifices accuracy.

Final Summary

This project demonstrates end-to-end regression modeling for real-world environmental data. From preprocessing and feature engineering to model comparison and hyperparameter tuning, the workflow provides a strong foundation for predictive environmental analytics.